

Uncoupled solution of the Navier–Stokes–Brinkman equations

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Why uncoupling?

When discretised, the NSB equations lead to a system of $\approx 4N$ unknowns, where N is the number of “discretization points” (in our applications they could be $O(10^9)$).

- In 2D $N \approx 10^5$: coefficient matrices can be stored and factored on a good laptop.
- In 3D $N \approx 10^7$: can still be managed by iterative methods, but the system is quite ill conditioned, no effective preconditioner has been found to date.
- In 3D $N \approx 10^8$ becomes prohibitive.

⇒ The **coupled** solution of the 3D NSB equations is **not feasible** as of today for large problems! (or, in any case, it is too slow)

What can we do?

The idea

Try to uncouple the system in four subsystems:

- 3 for the momentum equation
- one for the incompressibility constraint

Different approaches

We can divide the uncoupling approaches in two large categories:

- Exact uncoupling (usually very expensive)
- Approximate uncoupling (the so called fractional-step methods, we will concentrate on these)

Both approaches can be formulated either

- starting from the differential equations or
- starting from the discretised equations

Despite the claims of several authors, the two approaches are **essentially equivalent**.

In this lecture I will give you a brief introduction to the **fractional-step** methods.

Outline

- 1 Exact methods
 - Discrete methods
- 2 Fractional step methods
 - Fractional-step methods
 - Space-factored FSM
- 3 References

Basic concepts

The basic idea

Exact uncoupling methods are essentially based on the exact block factorization of the Navier–Stokes/Stokes operator.

- These methods rely on the linearity of the differential operator: can be applied to a linear problem only.
- Most uncoupling methods (all methods acting at the continuum level) hinge on the Stokes operator: discrete operators that can be built and factored once and for all at the beginning of the computation can be obtained.

Just to give you an idea, I will explain how these techniques work in a **discrete setting** (its simpler!)

The discrete problem

Let us start from the discretised equations.

When the Navier–Stokes equations are properly discretised, the discrete operator is block symmetric (G is the discrete gradient, G^T is the discrete divergence, A is the metaharmonic or advection-diffusion-reaction operator:

$$\begin{pmatrix} A & G \\ G^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{u} \\ p \end{pmatrix} = \begin{pmatrix} \mathbf{f} \\ 0 \end{pmatrix}.$$

A straightforward Gaussian elimination gives (A is nonsingular in general):

$$\begin{pmatrix} A & G \\ 0 & G^T A^{-1} G \end{pmatrix} \begin{pmatrix} \mathbf{u} \\ p \end{pmatrix} = \begin{pmatrix} \mathbf{f} \\ G^T A^{-1} \mathbf{f} \end{pmatrix}.$$

Then the pressure is first computed solving $G^T A^{-1} G p = G^T A^{-1} \mathbf{f}$, then velocity is computed from $A \mathbf{u} = \mathbf{f} - G p$.

Uzawa method

The procedure just illustrated seems very easy, but the pressure system matrix $G^T A^{-1} G$ is **full** and its computation is **prohibitive** for nontrivial problems.

Solution: the Uzawa method

A matrix-free, Krylov-subspace method can be used to solve the linear system: one has to compute the product of the matrix $G^T A^{-1} G$ by a given vector, this is accomplished in three steps: the product by G , the solution of a linear system with coefficient matrix A and the product by G^T .

- If A is symmetric (Laplacian or metaharmonic operator) a conjugate gradient solver can be employed.
- otherwise a method such as the GMRES is employed.

Preconditioning of the iterative solver is still an **open issue**.

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The need for approximate uncoupling

As we have seen in the previous section:

- Exact uncoupling methods are very expensive, especially in 3D
- They often rely on iterative solvers whose preconditioning is an open problem
- The use of efficient linear-system solvers such as multigrid solvers is not trivial

Exact uncoupling methods make sense if, for some reason, all this can be obtained at a reduced cost.

Otherwise, why should we bother with the exactness of uncoupling, when in any case we need to discretise the equations in time?

The basic idea

Let us consider the following system of ODEs $\frac{\partial w}{\partial t} + Lw = f$ that is discretized by the implicit Euler scheme

$$\frac{w^{n+1} - w^n}{\Delta t} + Lw^{n+1} = f$$

and suppose

- L can be expressed as $L = L_1 + L_2$
- L_1 and L_2 are “easier” than L

then we can approximate the time integration by:

$$\frac{w^* - w^n}{\Delta t} + L_1 w^* = f; \quad \frac{w^{n+1} - w^*}{\Delta t} + L_2 w^{n+1} = 0.$$

This makes sense if the cascade of equation can be solved more easily with respect to the original problem.

The error of the fractional-step method

We can analyse the error in the FSM as described in the previous slide

- 1 compute w^* from the second substep:

$$w^* = w^{n+1} + \Delta t L_2 w^{n+1}.$$

- 2 eliminate w^* in the first substep

$$\frac{w^{n+1} - w^n}{\Delta t} + L w^{n+1} = f + \Delta t L_1 (-L_2 w^{n+1}).$$

That shows that the error is of the order of Δt .

Chorin's projection method

Among fractional-step methods, projection methods have been formulated first.

They were first proposed by Alexandre Chorin, and leverage the Hodge decomposition. A theorem helps us:

Decomposition theorem

Any vector field $\mathbf{v}$ defined in Ω admits the unique orthogonal decomposition

$$\mathbf{v} = \mathbf{w} + \nabla\phi,$$

where $\mathbf{w}$ is a field solenoidal in Ω and with zero normal component on the boundary $\partial\Omega$ of Ω .

Chorin's projection method (2)

In Chorin's projection method two steps are performed:

- 1 a step to advance the momentum equation

$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} - \nu \nabla^2 \mathbf{u}^* + \frac{\nu}{k} \mathbf{u}^* = \mathbf{f}, \quad \mathbf{u}_{\partial\Omega}^* = 0.$$

- 2 a step to enforce the incompressibility

$$\begin{cases} \frac{\mathbf{u}^{n+1} - \mathbf{u}^*}{\Delta t} + \nabla p^{n+1} = 0, & \mathbf{u}^{n+1} \cdot \mathbf{n}_{\partial\Omega} = 0 \\ \nabla \cdot \mathbf{u}^{n+1} = 0 \end{cases}$$

Often the incompressibility step is rewritten as a Poisson equation for the pressure: by taking the divergence of the first equation and using the second one

$$\nabla^2 p^{n+1} = \nabla \cdot \mathbf{u}^* / \Delta t, \quad \partial p^{n+1} / \partial n|_{\partial\Omega} = 0 \rightarrow \mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla p^{n+1}.$$

More accurate schemes: pressure correction

Chorin's scheme is only first-order accurate for two reasons:

- ① a first-order backward Euler scheme has been employed
- ② but, **more important**, a first-order time splitting error affects this scheme.

If we use the old time step pressure in the first step we obtain a **pressure correction scheme**, with **second-order** time-splitting error. The first step is now:

$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} - \nu \nabla^2 \mathbf{u}^* + \frac{\nu}{k} \mathbf{u}^* = -\nabla p^n + \mathbf{f}, \quad \mathbf{u}^*_{\partial\Omega} = 0.$$

and the projection step, as a Poisson equation for pressure:

$$\nabla^2 \Delta p^{n+1} = \nabla \cdot \mathbf{u}^* / \Delta t, \quad \partial \Delta p^{n+1} / \partial n|_{\partial\Omega} = 0 \rightarrow \mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla \Delta p^{n+1}.$$

The difficulty in projection methods

The main difficulty in obtaining a fast, scalable solver based on the projection method is the solution of the Poisson equation for the pressure. There are several techniques to solve the corresponding linear system:

- Multigrid methods
- Methods based on the Fast Fourier Transform.

but

- Both do not allow nonuniform grids (important to resolve small features),
- FFT does not scale very well in parallel,
- multigrid is an iterative technique that requires iterations.

Can we do better?

A generalisation of pressure-correction schemes

Pressure correction schemes can be stated as an $O(\epsilon^2)$ perturbation of the INSE (cfr. Guermond, Mineev & Shen):

$$\begin{cases} \frac{\partial \mathbf{u}_\epsilon}{\partial t} - \nu \nabla^2 \mathbf{u}_\epsilon + \frac{\nu}{k} \mathbf{u}_\epsilon + \nabla p_\epsilon = \mathbf{f}, & \mathbf{u}_\epsilon|_{\partial\Omega} = 0 \\ -\epsilon \nabla^2 \phi_\epsilon + \nabla \cdot \mathbf{u}_\epsilon = 0 & \frac{\partial \phi_\epsilon}{\partial n} = 0, \\ \epsilon \frac{\partial p_\epsilon}{\partial t} = \phi_\epsilon - \chi \nu \nabla \cdot \mathbf{u}_\epsilon. \end{cases}$$

$\epsilon = \Delta t$. $\mathbf{u}_\epsilon$ and p_ϵ still converge to $\mathbf{u}$ and p as $\epsilon \rightarrow 0$ if we replace the Laplacian in $\nabla^2 \phi_\epsilon$ by a more general operator A , provided it enjoys the following properties: $a(p, q) = \int_\Omega q A p$ is symmetric and

$$\|q\|_2^2 \leq a(q, q), \forall q \in D(A).$$

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An efficient scheme based on direction splitting

An interesting operator A satisfying the aforementioned properties is the following

$$A = -(1 - \partial_{xx})(1 - \partial_{yy})(1 - \partial_{zz}).$$

This operator is interesting since solving $Ap = f$ amounts to the solution of a cascade of three 1D equations:

$$\begin{aligned}\psi - \partial_{xx}\psi &= -\frac{1}{\Delta t} \nabla \cdot \mathbf{u}^{n+1}, & \partial_x \psi|_{x=0,1} &= 0; \\ \varphi - \partial_{yy}\varphi &= \psi, & \partial_y \varphi|_{y=0,1} &= 0; \\ \phi^{n+1/2} - \partial_{zz}\phi^{n+1/2} &= \varphi, & \partial_z \phi|_{z=0,1} &= 0.\end{aligned}$$

which requires just $O(N^3)$ operations, in 3D $\rightarrow$ **optimal computational complexity**.

Crank-Nicolson discretisation of the viscous step

In order to obtain a second order scheme, we use Crank-Nicolson for the time discretisation. The momentum equation reads:

$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} + \left(\frac{\nu}{2k} - \frac{\nu}{2} \nabla^2 \right) (\mathbf{u}^* + \mathbf{u}^n) = \mathbf{f}^{n+1/2} - \nabla p_\star^{n+1/2}$$

where $p_\star^{k+1/2} = p^{k-1/2} + \phi^{k-1/2}$. Subtracting and summing $\mathbf{u}^n$ in the second term,

$$\begin{aligned} & \frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} + \left(\frac{\nu}{2k} - \frac{\nu}{2} \nabla^2 \right) (\mathbf{u}^* - \mathbf{u}^n) \\ &= \mathbf{f}^{n+1/2} - \nabla p_\star^{n+1/2} - \left(\frac{\nu}{k} - \nu \nabla^2 \right) \mathbf{u}^n \equiv \mathbf{g}^{n+1/2} \end{aligned}$$

that rewrites

$$\left(1 + \frac{\Delta t \nu}{2k} - \frac{\Delta t \nu}{2} \nabla^2 \right) (\mathbf{u}^* - \mathbf{u}^n) = \Delta t \mathbf{g}^{n+1/2}$$

Direction splitting for the viscous step

In 1962 Douglas proposed a direction splitting for the heat equation. We will use a slight modification that is thought to be unconditionally stable for our problem (see the papers by Angot, Keating and Mineev, 2012 and Gornak, Guermond, Iliev and Mineev, 2013). We can write the previous expression as:

$$(1 - \gamma \nabla^2)(\mathbf{u}^* - \mathbf{u}^n) = \frac{\Delta t}{\beta} \mathbf{g}^{n+1/2}$$

where $\gamma = \frac{\Delta t \nu}{2\beta}$ and $\beta = 1 + \frac{\Delta t \nu}{2k}$. We can substitute the operator $1 - \gamma \nabla^2$ with a second-order approximation:

$$\begin{aligned} (1 - \gamma \nabla^2) &\approx (1 - \gamma \partial_{xx})(1 - \gamma \partial_{yy})(1 - \gamma \partial_{zz}) \\ &= (1 - \gamma \nabla^2) + \gamma^2 (\partial_{xxyy} + \partial_{xxzz} + \partial_{yyzz}) - \gamma^3 \partial_{xxyyzz} \end{aligned}$$

which allows also the velocity to be advanced in $O(N^3)$ operations.

Direction splitting for the viscous step

The modified direction-splitting time step for the momentum equation reads:

$$\xi^{n+1} - \mathbf{u}^n = \frac{\Delta t}{\beta} \hat{\mathbf{g}}^{n+1/2}$$

$$\eta^{n+1} - \xi^{n+1} - \gamma \partial_{xx}(\eta^{n+1} - \eta^n) = 0$$

$$\zeta^{n+1} - \eta^{n+1} - \gamma \partial_{yy}(\zeta^{n+1} - \zeta^n) = 0$$

$$\mathbf{u}^{n+1} - \zeta^{n+1} - \gamma \partial_{zz}(\mathbf{u}^{n+1} - \mathbf{u}^n) = 0$$

which allows also the velocity to be advanced in $O(N^3)$ operations.
Here

$$\hat{\mathbf{g}}^{n+1/2} = \mathbf{f}^{n+1/2} - \nabla p_\star^{n+1/2} - \frac{\nu}{2k} \mathbf{u}^n + \frac{\nu}{2} (\partial_{xx} \eta^n + \partial_{yy} \zeta^n + \partial_{zz} \mathbf{u}^n).$$

Solution algorithm

A four step solution algorithm can be formulated:

- 1 The pressure predictor is computed to calculate $\nabla p_{\star}^{n+1/2}$ and then the right-hand side of the momentum equation $\hat{\mathbf{g}}^{n+1/2}$ are computed;
- 2 The momentum equation is solved by the modified Douglas direction-splitting algorithm to compute $\mathbf{u}^{n+1}$
- 3 The pressure correction is computed using the direction splitting algorithm for the pressure;
- 4 The pressure is updated: $p^{n+1/2} = p^{n-1/2} + \phi^{n+1/2}$.

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Some references

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